

BDR THERMEA GROUP

Automatic Model Validation using Pipelines

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1 Introduction

In the realm of thermal systems, heat pumps stand out as a cornerstone of modern heating and cooling solutions. They are pivotal in transitioning towards energy-efficient and environmentally friendly technologies. The efficacy of these systems hinges on the precision and reliability of their underlying models. This is where model validation becomes vital.

Model validation in heat pumps is a rigorous process of ensuring that the computational models accurately represent the real-world behavior of these systems. It involves a meticulous comparison of the model's output with empirical data. The essence of this process lies in its ability to guarantee that the models predict the performance, efficiency, and environmental impact of heat pumps under various conditions with high fidelity.

In an era where sustainability and energy efficiency are paramount, the role of heat pumps is increasingly significant. They are not only a solution for reducing carbon footprints in residential and commercial spaces but also a key element in the larger tapestry of renewable energy systems. Thus, ensuring the accuracy and reliability of heat pump models through validation is not just a technical necessity; it's a step towards a sustainable future.

As we embark on this project with BDR Thermea, our focus will be on developing and refining methodologies for automatic model validation. This will not only streamline the process but also enhance the efficiency and accuracy of model validation in heat pumps. By doing so, we aim to contribute significantly to the development of more efficient, reliable, and sustainable heating and cooling systems.

2 Presentation of BDR Thermea

2.1 Company Overview

BDR Thermea is a renowned leader in the heating and cooling industry, known for its commitment to innovation and sustainability. With a rich history that spans

over decades, BDR Thermea has established itself as a key player in providing advanced heating solutions globally. The company's vision is centered around delivering energy-efficient and environmentally friendly heating technologies, reflecting its dedication to tackling modern energy challenges.

2.2 Missions of BDR Thermea

At the heart of BDR Thermea's operations are its core missions: to lead in the transition towards a carbon-neutral future, to innovate in the field of thermal comfort, and to deliver exceptional value to its customers. These missions drive the company's efforts in developing cutting-edge technologies that not only meet current heating and cooling needs but also anticipate future demands in a world increasingly aware of environmental impacts.

2.3 CAE and Modeling Department

The Computer-Aided Engineering (CAE) and Modeling Department at BDR Thermea plays a crucial role in the company's innovation pipeline. This department is instrumental in developing sophisticated models and simulations that are vital for the design and improvement of heating systems.

The department's mission is to drive excellence in engineering through advanced modeling and simulation techniques. Its activities include developing and validating thermal models, conducting performance analyses, and optimizing design processes for new and existing products. The team's expertise in CAE and modeling is a cornerstone in BDR Thermea's commitment to delivering state-of-the-art heating and cooling solutions.

3 Overview of Heat Pumps

3.1 Introduction to Heat Pumps

Heat pumps are energy-efficient devices used for heating and cooling buildings. They operate by transferring heat from a cooler space to a warmer space, using the princi-

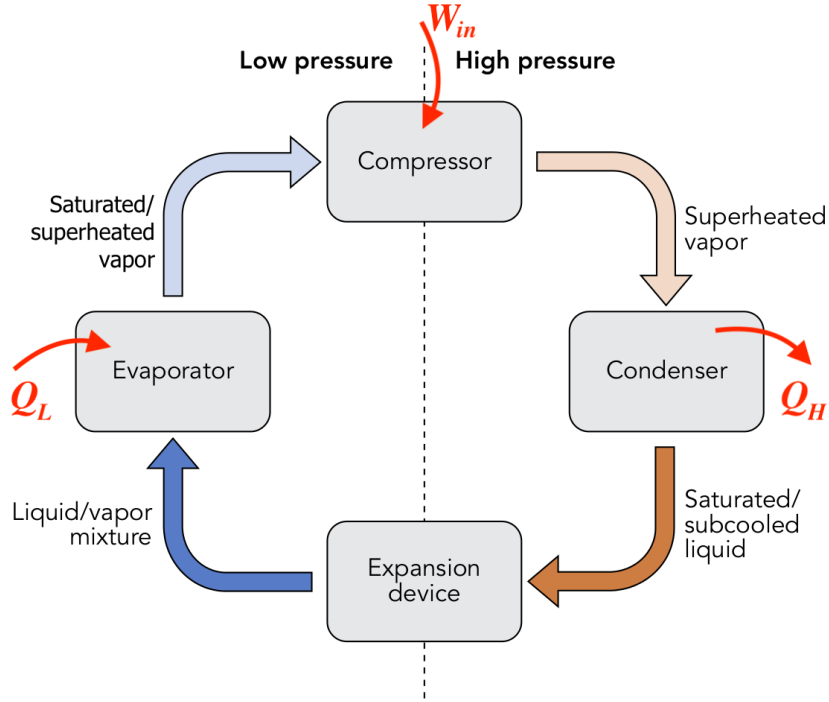
ples of thermodynamics. This process is reversible, allowing heat pumps to provide heating in winter and cooling in summer. They are considered environmentally friendly because they use less electrical energy compared to traditional heating and cooling systems.

3.2 Components of Heat Pumps

1. **Compressor:** The compressor is the heart of the heat pump system. It circulates the refrigerant through the system and increases the pressure and temperature of the refrigerant gas.
2. **Condenser:** In the condenser, the hot refrigerant gas releases its heat to the surroundings, condensing into a liquid. This heat is then used to warm the building or water in heating applications.
3. **Expansion Valve:** The expansion valve reduces the pressure of the refrigerant liquid, cooling it down rapidly. This is a critical component for regulating the flow of refrigerant and maintaining the efficiency of the system.
4. **Evaporator:** In the evaporator, the cold refrigerant absorbs heat from the external environment (air, ground, or water source). It then evaporates, turning into a low-pressure gas, and is sent back to the compressor to restart the cycle.

3.3 Working Principle

In heating mode, the heat pump extracts heat from external sources (like air, water, or ground) and transfers it indoors. In cooling mode, it reverses the process, removing heat from the interior and releasing it outside. This versatility makes heat pumps an increasingly popular choice for climate control in various settings.



4 Technical Components Presentation

In this section, we will explore the key technical components utilized in the project. These components form the backbone of our automatic model validation framework and are essential in ensuring the efficacy and accuracy of our methodologies.

4.1 Dymola

Dymola (Dynamic Modeling Laboratory) is a comprehensive tool for modeling and simulating complex integrated systems.

Dymola is used for developing detailed physical models using Modelica, a non-proprietary, object-oriented language designed for component-oriented modeling.

It allows for the creation of dynamic simulations of heat pump systems, enabling us to test various operational scenarios and compare the simulated performance with real-world data.

4.2 Thermal Systems

The Thermal Systems library is a crucial component of our project, serving as an integral part of the modeling and simulation environment within Dymola. This library is designed to provide a comprehensive set of tools and models for simulating thermal systems, particularly heat pumps.

The Thermal Systems library offers a wide range of tools and components that are specifically tailored for modeling various aspects of heat pump systems. This includes models for compressors, condensers, expansion valves and evaporators.

Utilizing the library, we develop detailed models of heat pump systems. These models are then used to simulate real-world scenarios, enabling us to analyze the performance of heat pumps under various conditions.

Models developed using the Thermal Systems library serve as benchmarks for validation. We compare these models' outputs with empirical data to ensure that they accurately represent the behavior of actual heat pump systems.

4.3 FMU/FMI

Functional Mock-up Interface (FMI) and Functional Mock-up Unit (FMU) are critical standards in the project:

FMI is an open standard for the exchange and co-simulation of dynamic models, while an FMU is a specific implementation of this standard.

They are used for integrating models developed in Dymola with other simulation tools and environments. This integration is essential for a comprehensive approach to model validation.

The use of FMI/FMU facilitates the consistent and accurate exchange of simulation data, ensuring that our models remain compatible and interoperable with different systems and tools, which is vital for a collaborative project like ours.

4.4 YAML Pipelines

YAML (YAML Ain't Markup Language) pipelines are an integral part of the automation process:

These pipelines are used to define the sequences of steps that the model validation process must follow, from data ingestion to simulation and analysis.

In our project, YAML pipelines are configured to automate various stages of model validation, ensuring consistency and efficiency.

5 Project Overview/Goal of the Project

5.1 Main Objectives

The project's main objectives are designed to streamline the model validation process by leveraging automation and integration techniques. These objectives include:

1. Development of an Automated Pipeline:

- The core task is to create a pipeline that can automatically invoke a software application on an agent pool. This software will execute models that are stored in a Git repository. The automation of this process is crucial for improving efficiency and reducing the potential for human error.

2. Extension of Pipeline Capabilities:

- Beyond basic model execution, the pipeline will be extended to perform more complex tasks. An example of this is the automatic generation of Functional Mock-up Units (FMUs), which are integral to the model validation process. This extension will demonstrate the pipeline's flexibility and capability to handle various aspects of model validation.

3. Automated Model Validation Methodology:

- A key deliverable is to propose and implement a method to automate the model validation process using pipelines. This involves setting up a system

that can automatically compare the model outputs with empirical data, ensuring that the model accurately represents real-world scenarios.

4. Deployment to Other Components:

- Finally, the developed methodology and pipeline will be adapted and deployed for other components within the heat pump system. This scalability is essential to ensure that the entire heat pump model, not just individual parts, is validated efficiently and accurately.

5.2 Expected Outcomes

By achieving these objectives, the project is expected to deliver:

- **Enhanced Efficiency:** Automated pipelines significantly reduce the time and effort required for model validation.
- **Increased Accuracy:** Automation minimizes human error, leading to more accurate model validation results.
- **Scalability:** The ability to deploy the method to other components demonstrates the scalability of the solution, making it adaptable to various model types and sizes.

6 Roadmap and Next Steps

1. Understanding Heat Pumps:

- Prior to commencing technical work, familiarization with the principles and functioning of heat pumps has been completed through a dedicated formation. This step was crucial to gain an in-depth understanding of the systems we are modeling and validating.

2. Analysis of Current Repositories:

- The next step involves a thorough analysis of existing repositories. This includes understanding how heat pumps are currently modeled, how these

models are stored, and the processes for calling these models. This stage is vital for identifying areas of improvement and integration for the automated pipeline.

3. Creating the Initial Pipeline:

- The primary task is to create a pipeline capable of calling software on an agent pool to execute models located in a Git repository. This step marks the beginning of the automation process in model validation.

4. Pipeline Extension:

- Following the initial setup, the pipeline will be extended to include more complex functionalities, such as automatic FMU generation. This extension is aimed at demonstrating the pipeline's adaptability and comprehensive coverage of the validation process.

5. Automating Model Validation:

- The project will explore various methodologies to automate the model validation process using pipelines. This involves creating a system that can automatically compare the model outputs with empirical data for accurate validation.

6. Deployment to Other Models:

- Finally, the validated method will be deployed across other models, ensuring that the automated validation process is not limited to a single model but is scalable and adaptable to various models within the heat pump system.

The final aim is to establish an automated, accurate, and scalable model validation methodology for heat pumps.

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